Xuanming Zhang

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Research Interests

Static, policy-based AI alignment fails to navigate the dynamic and often contradictory nature of human values. My research introduces Humanoid Intelligence as a new paradigm, aiming to build AI that intrinsically reasons about social context rather than merely following predefined patterns. The ultimate goal is to create AI that can deeply understand and carry forward human will, enabling future applications like the digital continuity of consciousness. My work is concentrated in these core areas:

- Cognitive Learning: Building trustworthy AI that can generalize its understanding of human values to novel situations. My work develops cognitive architectures including test-time learning mechanisms and metacognitive multi-agent system that enable case-by-case reasoning, moving beyond the limitations of fixed training data to achieve robust, adaptable alignment.
- World Interaction: Developing agents that learn through immersive interaction within simulated, long-horizon social scenarios. This allows for the study and shaping of complex emergent behaviors like deception, cooperation, and contextual decision-making in a controlled yet realistic setting.
- Policy Simulation: Conducting evaluation by forecasting the long-term societal impact of an agent's decisions. This shifts the measure of success from immediate, task-specific correctness to the broader, downstream consequences of AI behavior in complex systems.

Education

Visiting B. S. in Computer Science Jun. 2025 Aug. 2025
Stanford University, Stanford, CA
- Overall GPA: A+.

B. S. in Information Science Jan. 2025 May. 2026
University of Wisconsin-Madison, Madison, WI
- Overall GPA: 3.84/4.0.

B.S. in Artificial Intelligence Sept. 2021 Dec. 2024
Dropout from the Program of National Elite Institute of Engineering
- Research rotation phase in EECS, Peking University (Jun. 2023 Dec. 2024), advised by Prof. Zhi Jin, Ge Li.
  - Member of Key Laboratory of High Confidence Software Technologies (Peking University), Ministry of Education.
- General education phase in Chongqing University (Sept. 2021 May. 2023), advised by Prof. Zexiang Li.
  - CTO at startup being invested $70,000 under XbotPark.
  - Outstanding Student (Top 1), First-Class Scholarship.

Awards and Certificate

- National Level II Psychological Counselor, PCIA, 2025.
- The Chinese Mathematics Competitions for College Students, First Prize, 2023.
- National College Student Innovation and Entrepreneurship Training Program ($10,000), Ministry of Education in China, 2023.

Publications

See full list in my google scholar page, I publish under the name "Xuanming Zhang".

11. Cognition-of-Thought Elicits Social-Aligned Reasoning in Large Language Models
Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
International Conference on Learning Representations (ICLR) under review, 2026.
Advances in Neural Information Processing Systems (NeurIPS) 2025 Workshop on Socially Responsible and Trustworthy Foundation Models.
10. Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions
Yang Xu*, **Xuanming Zhang***, Min-Hsuan Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Yixuan Li (*Equal contribution)
International Conference on Learning Representations (ICLR) under review, 2026.
Advances in Neural Information Processing Systems (NeurIPS) 2025 Workshop on Socially Responsible and Trustworthy Foundation Models.
9. MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems
Xuanming Zhang, Yuxuan Chen, Min-Hsuan Yeh, Yixuan Li
Advances in Neural Information Processing Systems (NeurIPS) 2025, **spotlight**.
8. Human Decision-making is Susceptible to AI-driven Manipulation
Sahand Sabour, June M Liu, Siyang Liu, Chris Z Yao, Shiyao Cui, **Xuanming Zhang**, Wen Zhang, Yaru Cao, Advait Bhat, Jian Guan, Wei Wu, Rada Mihalcea, Hongning Wang, Tim Althoff, Tatia Lee, Minlie Huang
Nature Communications under revision, 2025.
7. Generalization or Memorization: Dynamic Decoding for Mode Steering
Xuanming Zhang
Advances in Neural Information Processing Systems (NeurIPS) 2025 Workshop on Socially Responsible and Trustworthy Foundation Models.
6. SocialEval: Evaluating Social Intelligence of Large Language Models
Jinfeng Zhou, Yuxuan Chen, Yihan Shi, **Xuanming Zhang**, Leqi Lei, Yi Feng, Zexuan Xiong, Miao Yan, Xunzhi Wang, Yaru Cao, Jianing Yin, Shuai Wang, Quanyu Dai, Zhenhua Dong, Hongning Wang, Minlie Huang
Annual Meeting of the Association for Computational Linguistics (ACL) 2025.
5. Seeker: Towards Exception Safety Code Generation with Intermediate Language Agents Framework
Xuanming Zhang, Yuxuan Chen, Yuan Yuan, Minlie Huang
Advances in Neural Information Processing Systems (NeurIPS) 2025 Workshop on Socially Responsible and Trustworthy Foundation Models.
4. Deep Graph Learning for Industrial Carbon Emission Analysis and Policy Impact
Xuanming Zhang

Advances in Neural Information Processing Systems (NeurIPS) 2025 Workshop AI for Science: The Reach and Limits of AI for Scientific Discovery.

3. EvoCodeBench: An Evolving Code Generation Benchmark with Domain-specific Evaluations
   Jia Li, Ge Li, **Xuanming Zhang**, Yunfei Zhao, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
   Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024.

2. DevEval: A Manually-Annotated Code Generation Benchmark Aligned with Real-World Code Repositories
   Jia Li, Ge Li, Yunfei Zhao, Yongmin Li, Huanyu Liu, Hao Zhu, Lecheng Wang, Kaibo Liu, Zheng Fang, Lanshen Wang, Jiazheng Ding, **Xuanming Zhang**, Yuqi Zhu, Yihong Dong, Zhi Jin, Binhua Li, Fei Huang, Yongbin Li
   Annual Meeting of the Association for Computational Linguistics (ACL) 2024.

Preprints

1. Theoretical Proof that Auto-regressive Language Models Collapse when Real-world Data is a Finite Set
   Lecheng Wang, Xianjie Shi, Ge Li, Jia Li, **Xuanming Zhang**, Yihong Dong, Wenpin Jiao, Hong Mei
   Technical Report

Service

Reviewer for

- NeurIPS (Top Reviewer), ICLR
- OOPSLA
- AAAI
- TMLR

Mentees

3. Yang Xu, undergraduate student at ZJU, 2025.
2. Tingting Du, undergraduate student at UW-Madison, 2025.
1. Yuxuan Chen, B.S. at THU, 2024, now MS student at THU.

Research Experience

Student Researcher May 2025 - Sep. 2025
Amazon AGI, Remote, USA

- Hosts: Dr. Rahul Gupta, Jwala Dhamala and Ousmane Dia.
- Research on *Trustworthy AI*.

Research Assistant Feb. 2025 - May. 2026
Department of Computer Sciences, UW-Madison, Madison, WI, USA

- Supervised by Dr. Sharon Li.
- Research on *Socially aligned LLMs*.

AI Architect Intern Sep. 2024 - Jan. 2025
Flow, Bytedance, Beijing, China

- Supervised by Dr. Xingnan Wang and Xuebin Lin.
- Research and design on *Coze agentic system*.

Research Scientist Intern Apr. 2024 - Jan. 2025
Department of Computer Science, Tsinghua University & Z.ai, Beijing, China

- Supervised by Dr. Minlie Huang.

- Research on *AI alignment, safety and ethics*.

Member Jun. 2023 - May. 2024

Key Lab of High Confidence Software Technologies, Ministry of Education &
School of Computer Science, Peking University, Beijing, China

- Supervised by Dr. Zhi Jin and Dr. Ge Li.
- Research on *Controlled-Text Generation*.

Additional Information

Language skills

- Native speakers of Mandarin with professional English speaking capability .

Programming skills

- Proficient with Python, PyTorch. Familiar with Matlab, C++, Swift.

Leadership skills

- I worked as CTO at the startup H3O from 2021-2022 and secured $70,000 in investment from XbotPark.
- I was the president of the student union both in BNDS and in college.